

CRM Lite Platform

Customer Follow-Up & Sales Lead Management System

A full-stack, enterprise-grade CRM engineered for modern sales pipelines, featuring real-time Kanban progression, atomic transaction customer conversion, automated follow-up scheduling, comprehensive chronological audit timelines, and live KPI intelligence dashboards.

Document Ref:	CRMLITE-TECH-SPEC-2026	Architecture:	Decoupled SPA / REST API
Backend Engine:	Python 3.14 / Django 6.1 / DRF 3.18	Database:	PostgreSQL (Prod) / SQLite (Dev)
Frontend Stack:	React 19 / Vite 8.3 / React Router v7	Authentication:	JWT (Access & Refresh Rotation)
Design System:	Vanilla CSS Glassmorphism + Dual Themes	API Docs:	OpenAPI 3.0 (drf-spectacular / Swagger)
Release Date:	September 2026	Author:	Engineering & Solutions Architecture Team

Pre-Configured Development & Evaluation Accounts

Role	Default Email / Username	Default Password	Scope & Core Capabilities
Admin / Mentor	admin@crmlite.com	Admin@123	Full control, team management, lead deletion, global reporting
Sales Manager	manager@crmlite.com	Manager@123	Team lead visibility, rep assignment, customer conversion, CSV reports
Sales Executive (Alex)	alex@crmlite.com	Alex@123	Assigned lead pipeline, follow-up scheduling, notes logging
Sales Executive (Sarah)	sarah@crmlite.com	Sarah@123	Assigned lead pipeline, follow-up scheduling, notes logging

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1. Executive Summary & System Overview

CRM Lite is a specialized, production-ready Customer Relationship Management application designed to streamline the sales pipeline from initial lead generation through customer conversion and post-sale retention. Traditional enterprise CRMs often suffer from bloated user interfaces, excessive configuration overhead, and poor performance. CRM Lite addresses these challenges by delivering an agile, high-performance platform centered on five core operational pillars:

- **Visual Kanban Pipeline:** Real-time deal progression through distinct pipeline stages (New, Contacted, Demo Scheduled, Negotiation, Qualified, Won, Lost) with immediate status transitions.
- **Atomic Customer Conversion:** Safe, transaction-wrapped conversion of qualified leads into permanent Customer accounts, preventing partial or duplicate records and preserving complete historical audit notes.
- **Proactive Follow-up Engine:** Smart scheduling of future interactions with automated overdue detection, today's schedule alerts, and mandatory outcome logging upon completion.
- **Chronological Communication Audit Trail:** Detailed logging of calls, WhatsApp messages, emails, demos, and objection notes linked to both reps and leads for full team transparency.
- **Live Real-Time KPI Analytics:** Direct database aggregation of win rates, pipeline value, monthly trends, and sales rep performance without hardcoded metrics or stale caches.

2. High-Level System Architecture

CRM Lite uses a decoupled, three-tier architecture separating the client single-page application (SPA), stateless RESTful backend services, and relational persistence storage. This separation ensures horizontal scalability, independent frontend and backend deployments, and security enforcement at the API gateway layer.

Layer	Technology	Responsibilities & Functional Scope
Client Presentation	React 19, Vite 8.3, React Router v7	Single Page Application (SPA), role-based view routing, glassmorphism UI, theme state context, Axios interceptor for JWT auto-refresh.
API Gateway / Proxy	Nginx / Caddy (Reverse Proxy)	SSL/TLS termination, static asset caching, request routing, rate limiting, and CORS header management.
Backend REST Core	Django 6.1, DRF 3.18	Stateless REST API, JWT authentication, RBAC permission enforcement, business validation rules, atomic database transactions, OpenAPI documentation.
Persistence Store	PostgreSQL 16+ (SQLite dev fallback)	ACID transactional relational data, foreign key integrity constraints (PROTECT/CASCADE), composite B-tree indexes on lookup columns.
Search Microservice	FastAPI, Uvicorn	Optional sub-10ms federated prospect search across leads and customers with in-memory caching.

End-to-End Sales Lead Lifecycle:

- 1. **Ingestion:** Lead created via web form, LinkedIn, or referral → Assigned status NEW.
- 2. **Assignment:** Sales Manager reviews lead profile and assigns to an available Sales Executive.
- 3. **Outreach & Follow-up:** Rep logs calls, demos, or emails; schedules next follow-up date and time.
- 4. **Pipeline Stages:** Deal moves through CONTACTED → DEMO_SCHEDULED → NEGOTIATION → QUALIFIED.
- 5. **Conversion or Loss:** If qualified, Sales Manager triggers /convert/ inside an atomic transaction, generating a Customer profile and marking the lead WON. If lost, a mandatory reason is recorded.

3. Role-Based Access Control (RBAC) & Security

Security in CRM Lite is enforced at the server level on every incoming HTTP request. Client-side route guards and conditional button rendering serve strictly as user experience enhancements; all permissions are verified inside Django REST Framework permission classes and model querysets.

Operation / Capability	Admin / Mentor	Sales Manager	Sales Executive
User Management (Create/Delete)	Full Control	View Only	No Access (403)
Lead Listing & Visibility	All Leads Org-Wide	All Team Leads	Assigned / Created Only
Create & Edit Leads	Yes (Global)	Yes (Team)	Yes (Own Leads)
Delete Lead Records	Yes (Permitted)	Forbidden (403)	Forbidden (403)
Assign / Reassign Leads	Yes (Any User)	Yes (Team Members)	Forbidden (403)
Convert Lead to Customer	Yes (Qualified Leads)	Yes (Qualified Leads)	Forbidden (403)
Communication Notes & Follow-ups	Yes (Global)	Yes (Team)	Yes (Own Leads Only)
Performance Reports	Full Organization	Full Team Summary	Own Metrics Scope
Export Reports to CSV	Yes (Unlimited)	Yes (Team Data)	Forbidden (403)

Authentication & Token Lifecycle Architecture:

- **JWT Token Pair:** On successful authentication (POST /api/auth/login/), the server returns a short-lived Access Token (60 min expiration) and a long-lived Refresh Token (7 days).
- **Silent Interceptor Refresh:** The frontend Axios client automatically intercepts 401 Unauthorized responses, queues concurrent requests, fetches a fresh access token from /api/auth/refresh/, and replays failed calls seamlessly.
- **Payload Claims:** User ID, email, role, and full name are embedded directly in the JWT payload for zero-roundtrip client permission checks.
- **Secure Password Hashing:** Passwords are protected using Django's PBKDF2 with SHA-256 algorithm (390,000 iterations by default).

4. Relational Database Schema & Data Models

The relational database schema is normalized to 3NF, utilizing strict foreign key constraints, one-to-one conversion mapping, and targeted indexes to guarantee sub-millisecond query performance.

Entity / Table	Key Columns & Types	Relationships & Constraints
User (accounts_user)	id (PK), email (UK), password, first_name, last_name, phone, role (Enum), is_active	Custom User model where email is the unique authentication field. Roles: ADMIN, MANAGER, EXECUTIVE.

LeadSource (leads_leadsource)	id (PK), name (UK), description, is_active, created_at	Tracks acquisition channels (Website, LinkedIn, Referral). Linked to leads via on_delete=models.SET_NULL.
Lead (leads_lead)	id (PK), name, phone (Index), email (Index), company_name, status (Index), priority, expected_value, lost_reason	Core prospect record. assigned_to FK to User. Active status prevents duplicate phone entries. Status: NEW, CONTACTED, DEMO, NEGOTIATION, QUALIFIED, WON, LOST.
Customer (customers_customer)	id (PK), lead_id (FK_UK), name, phone, email, company_name, address, converted_at	OneToOne relationship to Lead with on_delete=models.PROTECT to guarantee financial audit history cannot be deleted.
FollowUp (followups_followup)	id (PK), lead_id (FK), assigned_to_id (FK), follow_up_at (Index), purpose, status (Index), outcome	Follow-up tasks. Status: PENDING, COMPLETED, CANCELLED, OVERDUE. Past-date scheduling rejected by validator.
LeadNote (leads_leadnote)	id (PK), lead_id (FK), user_id (FK), note_type, note_text, created_at	Timeline communication entries. Types: CALL, WHATSAPP, EMAIL, MEETING, DEMO, OBJECTION, GENERAL.
ActivityLog (activity_activitylog)	id (PK), entity_type, entity_id, action, old_value (JSON), new_value (JSON), performed_by_id (FK)	Immutable audit trail logging state changes (stage transitions, assignments, customer conversions).

5. REST API Reference & Endpoints

All authenticated endpoints expect standard Bearer tokens: Authorization: Bearer <access_token>. The complete interactive Swagger UI is available at /api/docs/ and the OpenAPI schema is accessible at /api/schema/.

HTTP Method	Endpoint Path	Functionality & Query Parameters	Roles Permitted
POST	/api/auth/login/	Accepts email and password; returns JWT access/refresh pair and user object.	Public
POST	/api/auth/refresh/	Accepts refresh token; generates fresh access token.	Public
GET, PATCH	/api/auth/me/	Retrieves or updates authenticated user profile and contact details.	Authenticated
GET, POST	/api/leads/	List and create leads. Filter by status, priority, source, keyword.	All (Scoped)
GET, PATCH	/api/leads/{id}/	Retrieve and update single lead contact, status, or details.	Admin, Manager, Rep
DELETE	/api/leads/{id}/	Permanently remove lead record. Non-admins receive 403 Forbidden.	Admin Only
POST	/api/leads/{id}/convert/	Atomically convert QUALIFIED lead to Customer record and mark WON.	Admin, Manager
POST	/api/leads/{id}/assign/	Reassign lead to designated executive user ID.	Admin, Manager
GET, POST	/api/leads/{id}/notes/	List or record communication notes (Call, WhatsApp, Demo, etc.).	Admin, Manager, Rep
GET	/api/leads/{id}/timeline/	Chronological audit history of notes and system actions.	Admin, Manager, Rep
GET	/api/leads/pipeline/	Leads grouped into columns by pipeline stage for Kanban visualization.	All (Scoped)
GET, POST	/api/follow-ups/	List and schedule follow-ups. Reject past scheduling timestamps.	All (Scoped)
POST	/api/follow-ups/{id}/complete/	Mark task completed and persist required outcome details.	Assigned Rep, Mgr, Admin
GET	/api/follow-ups/overdue/	Retrieves all pending follow-ups where scheduled date < now.	All (Scoped)
GET	/api/customers/	List converted client accounts and conversion timestamps.	All (Read-Only)
GET	/api/reports/summary/	Real-time aggregated KPIs: conversion %, pipeline value, rep metrics.	All (Scoped)
GET	/api/reports/export/?format=csv	Streaming downloadable CSV export of filtered pipeline data.	Admin, Manager
GET, POST, PATCH	/api/users/	List, create, update, or deactivate team member accounts.	Admin, Manager (View)

6. Frontend Architecture & Design System

The user interface is engineered with React 19 and Vite 8.3 without heavy external CSS frameworks (such as Tailwind). Instead, it implements a bespoke Vanilla CSS design system utilizing CSS Custom Properties (CSS variables) and modern dark glassmorphism techniques with a native dual-theme engine (White & Emerald Green light mode, Deep Slate dark mode).

State Management & Context Layering:

The React application maintains clean, isolated concern boundaries through three top-level Context providers:

- **ThemeProvider (ThemeContext.jsx):** Governs dark/light state, sets the data-theme root attribute on the DOM, and synchronizes preference to localStorage.
- **AuthProvider (AuthContext.jsx):** Stores the active user profile, provides login() and logout() methods, and initializes user state from persisted local tokens.
- **ToastProvider (ToastContext.jsx):** Renders non-intrusive floating feedback notifications across all pages with automated timeout dismissals.

Key User Interface Views:

- **Dashboard (/dashboard):** Real-time KPI metric cards (Total Leads, Conversion Rate %, Pipeline Value, Overdue Tasks), monthly acquisition trend charts, and urgent follow-up tables.
- **Kanban Pipeline Board (/pipeline):** Interactive drag-and-drop board with edge auto-scrolling and cross-page funnel flipping. Dragging cards near the screen edge or hovering over stage tabs automatically navigates between stage groups (Active Funnel ↔ Closing & Won), while smooth horizontal auto-scrolling enables seamless card repositioning in All-Stages view.
- **Lead Management (/leads):** Data table with multi-parameter search, server pagination, priority badges, source chips, and quick action conversion modals.
- **Lead Detail & Audit Trail (/leads/:id):** Comprehensive 360-degree prospect dossier containing contact info, scheduled interactions, and a chronological communication audit timeline.
- **Follow-up Scheduler (/follow-ups):** Overdue alerts, today's schedule calendar, and quick-action completion drawers requiring logged interaction outcomes.
- **Customer Portfolio (/customers):** Permanent customer account directory with conversion timestamps and associated converted lead references.

7. Core Business Logic & Data Validation Rules

Atomic Customer Conversion: Conversion is wrapped inside `django.db.transaction.atomic()`. The lead status MUST be QUALIFIED. On conversion, a new Customer row is created, the lead status changes to WON, the converted_at timestamp is frozen, and an immutable ActivityLog is persisted. If any step fails, the entire transaction rolls back to prevent orphan records.

Phone Number Deduplication: Creating a lead with a phone number that already exists on an active prospect (status not in ['WON', 'LOST']) returns HTTP 400 Bad Request. This prevents multiple reps from unknowingly colliding on the same account.

Past-Date Follow-up Prevention: The FollowUpSerializer strictly validates that scheduled times cannot be in the past (`follow_up_at >= timezone.now() - timedelta(minutes=5)`). Any past timestamp is rejected with validation errors.

Mandatory Lost Reason: A lead cannot transition to status = 'LOST' with an empty `lost_reason` string. This guarantees that sales analytics can accurately aggregate win/loss root causes.

Negative Value Prevention: Expected deal revenue cannot be negative (`expected_value >= 0`). Enforced via model validator constraints.

8. Testing Strategy & Quality Assurance

CRM Lite includes automated test suites covering authentication, role isolation, lead CRUD, past follow-up rejection, and atomic customer conversion running on Django's test runner in isolated in-memory test databases.

Executing Backend Automated Tests:

```
python backend/manage.py test accounts leads followups
```

App / Module	Test Focus Area	Assertion & Verification Goal
accounts	JWT Auth & Profile	Verifies 200 on valid credentials; structured 401 on bad password; 401 on missing Authorization header; profile match on /api/auth/me/.
leads	Role Isolation	Verifies Sales Executive cannot view other reps' leads in queries; direct GET of other reps' lead returns 403/404; Manager sees full team.
leads	Lead Assignment	Verifies Manager can assign reps; Executive assignment returns 403 Forbidden.
leads	Data Integrity	Duplicate active phone numbers return 400 Bad Request; negative expected values rejected; missing lost reasons on LOST status return 400.
leads	Atomic Conversion	Qualified lead converts cleanly into Customer record; lead status updates to WON; duplicate conversion attempts return 400.
followups	Date & Outcome	Past scheduling rejected; future scheduling accepted; completion records outcome string and completed_at timestamp.
reports	Aggregation & CSV	Dashboard returns exact counts matching database; CSV export streams valid CSV headers with attachment Content-Disposition.

9. Production Deployment & DevOps

1. PostgreSQL Database Configuration: Create a dedicated user and database with full schema privileges:

```
CREATE DATABASE crm_lite_db;
CREATE USER crm_admin WITH PASSWORD 'StrongPassword123';
GRANT ALL PRIVILEGES ON DATABASE crm_lite_db TO crm_admin;
```

2. Environment Setup (backend/.env):

Set DEBUG=False, generate a unique SECRET_KEY, configure DB_ENGINE=postgresql, and restrict ALLOWED_HOSTS and CORS_ALLOWED_ORIGINS to your production domain.

3. Apply Migrations & Collect Static Files:

```
python backend/manage.py migrate
python backend/manage.py collectstatic --noinput
```

4. WSGI Production Server (Gunicorn):

```
gunicorn config.wsgi:application --chdir backend --bind 127.0.0.1:8000 --workers 3 --timeout 120
```

5. Frontend Static Asset Build:

cd frontend && npm run build (outputs optimized production bundle to dist/ for Nginx serving).

10. Troubleshooting & Common Resolutions

- **Issue: Login returning HTTP 400 Bad Request**

Resolution: Ensure client sends JSON payload with keys email and password. The backend serializer strictly requires email rather than a generic username field.

- **Issue: White Screen on Navigation (useTheme must be used within a ThemeProvider)**

Resolution: Ensure App.jsx wraps the component router tree inside <ThemeProvider>. Components like Navbar rely on theme state context.

- **Issue: CORS Errors in Browser Console**

Resolution: Verify that the frontend origin (e.g. http://localhost:5173) is listed in CORS_ALLOWED_ORIGINS in backend/.env.

- **Issue: Database Reset / Fresh Seed Data**

Resolution: To restore all default evaluation users and pipeline mock records, execute: python backend/manage.py seed_data.

End of Technical Specification — CRM Lite Platform